ADAPTIVE COGNITIVE & EDUCATIONAL SKILL SUPPORT (ACESS)
AUTHORITATIVE REFERENCES & BEST-PRACTICE GUIDE

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UNIVERSITI TEKNIKAL MALAYSIA MELAKA

BORANG PENGESAHAN STATUS LAPORAN

JUDUL :

ADAPTIVE COGNITIVE & EDUCATIONAL SKILL SUPPORT (ACESS) AUTHORITATIVE REFERENCES & BEST-PRACTICE GUIDE

SESI PENGAJIAN : 2024/2025

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ADAPTIVE COGNITIVE & EDUCATIONAL SKILL SUPPORT (ACESS)
AUTHORITATIVE REFERENCES & BEST-PRACTICE GUIDE

MUHAMMAD ALIFF BIN AFFANDI

This report is submitted in partial fulfillment of the requirements for the Bachelor of Computer Science (Software Engineering) with Honours.

FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY
UNIVERSITI TEKNIKAL MALAYSIA MELAKA

2026

DECLARATION

I hereby declare that this project entitled
ADAPTIVE COGNITIVE & EDUCATIONAL SKILL SUPPORT (ACESS): AUTHORITATIVE REFERENCES & BEST-PRACTICE GUIDE
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I hereby declare that I have read this project report and found this project report is sufficient in term of the scope and quality for the award of Bachelor of Computer Science (Software Engineering) with Honours.

SUPERVISOR : ______________________ Date : __________
(DR. NORHAZWANI BINTI MD YUNOS)

DEDICATION

This work is dedicated to my beloved parents, Fadli bin Maarof and Norizan binti Kassim, whose love, sacrifices, and unwavering belief in me have been my greatest source of strength throughout this journey.

To my friends, who stood by me through every challenge, offered a listening ear when I needed it most, and kept me going when things felt impossible — thank you for being there.

And to everyone who has supported me, in any way, big or small — this is as much yours as it is mine.

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I would also like to extend my sincere appreciation to my friend, Ahmad Mirza Shahmi bin Abdul Hanif, whose companionship and willingness to engage in meaningful discussions about this project have been a great source of motivation and encouragement. Having someone who truly understands the challenges of this work made the journey significantly more bearable.

Above all, I am profoundly grateful to my beloved parents, Fadli bin Maarof and Norizan binti Kassim, for their unconditional love, endless patience, and unwavering support throughout my academic journey. Their sacrifices and constant encouragement have been the foundation upon which I stand, and this achievement is as much theirs as it is mine.

ABSTRACT

This study develops and evaluates a machine learning framework for the automated classification of five transparent conductive oxide (TCO) materials — AZO, FTO, ITO, MZO, and ZnO — using their X-ray diffraction (XRD) structural signatures. TCOs are widely used in solar cells, OLEDs, LCDs, smart windows, and other optoelectronic devices, yet rapid and accurate phase identification still relies on slow, expert-dependent manual analysis of diffraction patterns. To address this, raw 2θ–intensity scans were converted into a rich set of physically meaningful features, including d-spacing descriptors derived from Bragg's Law, peak positions and widths, crystallite size, binned intensity profiles, and global statistical measures. Because experimental scans were limited, a two-strategy data augmentation pipeline combining physics-motivated perturbations and JCPDS-based simulation was used to expand the dataset to 640 samples. Fifteen supervised classifiers were implemented, each wrapped in a scikit-learn pipeline with feature scaling applied inside every cross-validation fold to prevent data leakage. Models were assessed using an 80/20 holdout split and repeated 5×5 stratified cross-validation, with accuracy, Cohen's kappa, log loss, macro F1-score, ROC-AUC, confusion matrices, and learning curves. LightGBM achieved the strongest performance, with a cross-validation accuracy of 89.9% (±2.1%) and a holdout accuracy of 90.6% (Cohen's kappa 0.88), followed closely by XGBoost, Gradient Boosting, and Linear Discriminant Analysis. A stacking ensemble of the top five models matched the best single model at 90.6% accuracy, and mean one-vs-rest ROC-AUC values reached 0.97–0.99. Feature importance analysis confirmed that intensity-window and d-spacing-based crystallographic features were the most discriminative, while the AZO class proved the most difficult to separate from the other wurtzite-structured materials. These results demonstrate that XRD structural signatures provide an accurate, instrument-independent basis for automated TCO identification, complementing prior optical-signature approaches and offering a reproducible tool for data-driven materials characterisation and industrial quality control.

ABSTRAK

Kajian ini membangunkan dan menilai satu rangka kerja pembelajaran mesin untuk pengelasan automatik lima bahan oksida konduktif lutsinar (TCO) — AZO, FTO, ITO, MZO, dan ZnO — menggunakan tanda struktur pembelauan sinar-X (XRD). Bahan TCO digunakan secara meluas dalam sel suria, paparan, dan peranti optoelektronik lain, namun pengenalpastian fasa yang pantas dan tepat masih bergantung pada analisis manual corak pembelauan yang lambat dan memerlukan kepakaran. Bagi menangani masalah ini, imbasan mentah 2θ–keamatan ditukarkan kepada satu set ciri bermakna secara fizikal, termasuk penghurai jarak-d yang diterbitkan daripada Hukum Bragg, kedudukan dan lebar puncak, saiz hablur, profil keamatan terbin, dan ukuran statistik global. Oleh kerana imbasan eksperimen adalah terhad, satu saluran penambahan data dua strategi yang menggabungkan gangguan bermotifkan fizik dan simulasi berasaskan JCPDS digunakan untuk mengembangkan set data kepada 640 sampel. Lima belas pengelas terselia dilaksanakan, setiap satunya dibungkus dalam saluran scikit-learn dengan penskalaan ciri dilakukan di dalam setiap lipatan pengesahan silang untuk mengelakkan kebocoran data. Model dinilai menggunakan pembahagian holdout 80/20 dan pengesahan silang berstrata 5×5 berulang, dengan ketepatan, kappa Cohen, kehilangan log, skor makro F1, ROC-AUC, matriks kekeliruan, dan lengkung pembelajaran. LightGBM mencapai prestasi terbaik, dengan ketepatan pengesahan silang 89.9% (±2.1%) dan ketepatan holdout 90.6% (kappa Cohen 0.88), diikuti rapat oleh XGBoost, Gradient Boosting, dan Analisis Diskriminan Linear. Ensembel penyusunan (stacking) lima model terbaik menyamai model tunggal terbaik pada ketepatan 90.6%, dan nilai purata ROC-AUC satu-lawan-semua mencapai 0.97–0.99. Analisis kepentingan ciri mengesahkan bahawa ciri tetingkap keamatan dan ciri berasaskan jarak-d merupakan ciri paling membezakan, manakala kelas AZO didapati paling sukar dibezakan daripada bahan berstruktur wurtzit yang lain. Keputusan ini menunjukkan bahawa tanda struktur XRD menyediakan asas yang tepat dan bebas-instrumen untuk pengenalpastian TCO automatik, melengkapkan pendekatan tanda optik terdahulu serta menawarkan alat boleh-ulang bagi pencirian bahan dipacu data dan kawalan kualiti industri.

TABLE OF CONTENTS

PAGE

LIST OF TABLES

PAGE

LIST OF FIGURES

PAGE

LIST OF ABBREVIATIONS

AZO	-	Aluminium-doped Zinc Oxide
AUC	-	Area Under the Curve
CNN	-	Convolutional Neural Network
CV	-	Cross-Validation
FTO	-	Fluorine-doped Tin Oxide
FWHM	-	Full Width at Half Maximum
FYP	-	Final Year Project
ITO	-	Indium Tin Oxide
JCPDS	-	Joint Committee on Powder Diffraction Standards
KNN	-	K-Nearest Neighbours
LCD	-	Liquid Crystal Display
LDA	-	Linear Discriminant Analysis
LightGBM	-	Light Gradient Boosting Machine
MLP	-	Multi-Layer Perceptron
ML	-	Machine Learning
MZO	-	Magnesium-doped Zinc Oxide
OLED	-	Organic Light-Emitting Diode
PSM	-	Projek Sarjana Muda
RF	-	Random Forest
ROC	-	Receiver Operating Characteristic
SHAP	-	SHapley Additive exPlanations
SVM	-	Support Vector Machine
TCO	-	Transparent Conductive Oxide
XAI	-	Explainable Software Engineering
XGBoost	-	Extreme Gradient Boosting
XRD	-	X-Ray Diffraction
ZnO	-	Zinc Oxide

LIST OF ATTACHMENTS

Guide]ACESS System — Authoritative References & Best-Practice

Guide

Adaptive Cognitive & Educational Skill Support (ACESS)

Compiled reference document covering technical stack, H5P, Moodle, accessibility standards, adaptive learning, LMS patterns, and academic research.

References]Section A: Technical Stack

References

Components]A1. Next.js App Router — Server vs Client

Components

Title: Next.js App Router — Official Documentation

Organization: Vercel / Next.js Team **Year:** 2024–2025 (continuously updated) **URL:** https://nextjs.org/docs/app

Summary: The App Router uses React Server Components (RSC), Suspense, and Server Functions. All components default to server-rendered; only those marked

`'use client'` ship JavaScript

to the browser, enabling selective hydration. **ACESS relevance:** Underpins ACESS's page architecture — server components for course listings, learner profiles, and data fetching; client components only where interactivity is needed (editors, dashboards, real-time progress bars).

Title: Getting Started: Server and Client Components

Organization: Vercel / Next.js Team **Year:** 2025

URL:

https://nextjs.org/docs/app/getting-started/server-and-client-components

Summary: Defines when to use each component type: server for data access and layouts, client (`'use client'`) for state, event handlers, and browser APIs. The mental model is "server-first, client-islands." **ACESS relevance:** Directly guides ACESS component design — lesson content renders on the server, interactive quiz and editor components run on the client.

Utility-First]A2. Tailwind CSS v4 — Theming &

Utility-First

Title: Tailwind CSS v4.0 — Official Blog Announcement

Organization: Tailwind Labs **Year:** 2025 **URL:**

https://tailwindcss.com/blog/tailwindcss-v4 **Summary:** v4 is a ground-up rewrite introducing CSS-first configuration via the `@theme` directive, a new Oxide engine (3.5× faster full builds, 100× faster incremental), native cascade layers, and wide-gamut P3 OKLCH color palettes. No more `tailwind.config.js` is needed.

ACESS relevance: ACESS's multi-theme accessibility profiles (dyslexia-friendly cream palette, high-contrast, dark mode) are all defined as `@theme` CSS variables, keeping every theme override in one CSS file with no JS config.

Title: Theme Variables — Core Concepts **Organization:**

Tailwind Labs **Year:** 2025 **URL:**

https://tailwindcss.com/docs/theme **Summary:** Explains how `@theme` CSS variables automatically generate utility classes. Defining `--color-primary-500` creates a `bg-primary-500` utility without extra config. Distinguishes `@theme` (generates utilities) from `:root` (raw CSS variables). **ACESS relevance:** ACESS accessibility profiles swap `--color-background`, `--font-family`, and `--line-height` theme variables dynamically, instantly reflecting throughout all components without component code changes.

Pattern]A3. shadcn/ui — Radix + CVA + tailwind-merge Pattern

Title: shadcn/ui — Official Component Library

Organization: shadcn **Year:** 2024–2025 (continuously updated) **URL:** https://ui.shadcn.com/ **Summary:** A collection of copy-paste React components built on Radix UI primitives and styled with Tailwind CSS. Instead of a published npm package, component source code is owned by the developer. CVA (class-variance-authority) manages variant styling; `cn()` via tailwind-merge handles class conflicts. **ACESS relevance:** All ACESS UI components (buttons, dialogs, tabs, dropdowns, cards) are built on shadcn/ui. Owning the component code means ACESS can adjust any component for accessibility or brand compliance without fighting a black-box library.

Title: The Anatomy of shadcn/ui Components

Organization: Manu Arora (independent technical breakdown)

Year: 2023 **URL:**

https://manupa.dev/blog/anatomy-of-shadcn-ui **Summary:**

Deep-dives the three-layer architecture: Radix UI for behavior/accessibility semantics, Tailwind CSS for visual styling, and CVA (`class-variance-authority`) for variant management. Shows how `badgeVariants = cva(...)` separates styling from rendering. **ACESS relevance:** Explains the exact pattern used throughout ACESS's `components/ui/` directory — every component variant (e.g., dyslexia mode, high-contrast) is a CVA variant, not a separate component.

Primitives]A4. Radix UI — WAI-ARIA Accessibility

Primitives

Title: Radix Primitives — Accessibility Overview

Organization: WorkOS / Radix UI **Year:** 2024 (continuously updated) **URL:** https://www.radix-ui.com/primitives/docs/overview/accessibility

Summary: Radix Primitives follow WAI-ARIA authoring guidelines and are tested across major browsers and assistive technologies. They handle aria/role attributes, focus management, focus trapping, and keyboard navigation automatically for Dialog, Tabs, Accordion, DropdownMenu, Popover, and more. **ACESS relevance:** Every ACESS modal, dropdown, and navigation component inherits WAI-ARIA compliance from Radix without manual ARIA attribute writing, critical for learners using screen readers.

Title: Radix Primitives — Introduction **Organization:** WorkOS / Radix UI **Year:** 2024 **URL:** https://www.radix-ui.com/primitives/docs/overview/introduction

Summary: Explains the goal: provide headless, unstyled components for common UI patterns (Accordion, Dialog, Combobox, Tabs, Slider, Select) that are inaccessible or missing in native HTML. Each component is independently versioned. **ACESS relevance:** ACESS uses Radix Dialog for lesson modals, Radix Tabs for course navigation, and Radix Accordion for collapsible settings — all receiving keyboard and screen reader support out of the box.

Title: Radix Primitives — Dialog Component

Organization: WorkOS / Radix UI **Year:** 2024

URL: https://www.radix-ui.com/primitives/docs/components/dialog

Summary: Adheres to the Dialog WAI-ARIA design pattern. Automatically traps focus within the dialog, manages `aria-describedby`, and closes on Escape. Supports controlled and uncontrolled modes. **ACESS relevance:** ACESS uses Dialog for settings panels, lesson completions, and confirmation dialogs. Focus trapping is critical for learners using keyboard or screen reader navigation.

Extensibility]A5. Tiptap — Rich Text Editor

Extensibility

Title: Tiptap Editor — Get Started **Organization:**

Überdosis (Tiptap) **Year:** 2025 (continuously updated)

URL: https://tiptap.dev/docs/editor/getting-started/overview

Summary: Tiptap is a headless rich-text editor framework built on ProseMirror. It uses an extension-based architecture — every feature (bold, image, table, mentions, custom nodes) is an extension. Works with React, Vue, Svelte, and plain JS. **ACESS relevance:** ACESS educators use Tiptap to author lesson content. Its extension model allows ACESS to add custom blocks (e.g., dyslexia-mode font toggle, embedded H5P, read-aloud buttons) without forking the editor.

Title: Tiptap React Integration **Organization:**

Überdosis (Tiptap) **Year:** 2025 **URL:**

https://tiptap.dev/docs/editor/getting-started/install/react

Summary: Shows the `useEditor` hook pattern, extension array, and declarative `<Tiptap>` component. Notes that `immediatelyRender: false` must be set for SSR (Next.js) compatibility to avoid hydration mismatches.

relevance:**ACESS**

relevance: Essential for ACESS's Next.js-based lesson editor — the

SSR flag prevents hydration errors; `EditorContext` **shares the**

editor state across toolbar, bubble menu, and lesson-save components.

Library]A6. Motion (Framer Motion) — React Animation

Library

Title: Motion for React — Get Started Organization:

Motion (formerly Framer Motion) Year: 2025 URL:

https://motion.dev/docs/react Summary: Motion for React

(rebranded from Framer Motion in mid-2025, now imported from

`motion/react`**) is a React animation library using a hybrid**

engine: Web Animations API for 120fps performance, with JavaScript

fallback for spring physics and gestures. Trusted by Framer and Figma.

ACESS relevance: ACESS uses Motion for page transition

animations, lesson progress celebrations, and UI microinteractions. All

imports updated to `motion/react`.

Title: Creating Accessible Animations in React — Motion Guide

Organization: Motion (Framer Motion) Year: 2025

URL: https://motion.dev/docs/react-accessibility

Summary: Documents the `reducedMotion="user"` prop on `MotionConfig` (disables transform/layout animations site-wide respecting OS settings) and the `useReducedMotion()` hook for per-component control. `reducedMotion="user"` preserves opacity/color transitions while disabling movement.

relevance:ACESS

relevance: ACESS's Distraction-Free Mode and ADHD/ASD accommodation

profiles use `reducedMotion="always"` via MotionConfig. The `useReducedMotion()` hook is used per-component to replace slide animations with fades.

SSR]A7. Supabase — Real-time, Row Level Security,

SSR

Title: Row Level Security — Supabase Docs

Organization: Supabase Year: 2025 (continuously

updated) URL:

https://supabase.com/docs/guides/database/postgres/row-level-security

Summary: Documents PostgreSQL RLS policies as "implicit WHERE

clauses." When RLS is enabled on a table, all queries are filtered by

policies using `auth.uid()`**. The anon key is public; RLS is the**

only barrier between the key and unauthorized data access.

relevance:ACESS

relevance: ACESS's learner data isolation — progress records,

accessibility profiles, notes — relies on RLS policies ensuring

learners see only their own rows and educators see only their enrolled

students' data.

Title: Realtime Authorization — Supabase Docs

Organization: Supabase Year: 2025 URL:

https://supabase.com/docs/guides/realtime/authorization

Summary: Realtime Broadcast and Presence channels are secured via RLS policies on the `realtime.messages` table. Users only receive real-time updates for rows their SELECT policy permits. Authorization is evaluated when the WebSocket connection is established.

ACESS relevance: ACESS's live class features (educator sees learner progress in real time, leaderboard updates) use Supabase Realtime with RLS ensuring cross-learner data never leaks through subscriptions.

Visualization]A8. Recharts — Accessible Data Visualization

Title: Recharts — Composable React Charting Library

Organization: Recharts Year: 2024 URL: https://recharts.org/ Summary: A lightweight SVG-based charting library built on D3 submodules, offering composable components (`<LineChart>`, `<BarChart>`,

`<PieChart>`,

`<ResponsiveContainer>`) that integrate

naturally with React's component model. Used by shadcn/ui's chart

components. ACESS relevance: ACESS educator and admin

dashboards use Recharts for learner progress charts, completion rates,

and module engagement analytics, all wrapped in shadcn/ui's chart

component layer.

Implementation]A9. next-themes — Dark Mode

Implementation

Title: next-themes — Perfect Next.js Dark Mode

Organization: Paco Coursey (pacocoursey) Year: 2024

URL: https://github.com/pacocoursey/next-themes

Summary: Injects a blocking script in

`<head>` before rendering to read the

user's stored theme preference from localStorage, preventing the

flash-of-incorrect-theme (FOIT) issue common in SSR. Wraps the app in a

`ThemeProvider` **and exposes** `useTheme()` **hook.**

ACESS relevance: ACESS uses next-themes to support light, dark, and custom accessibility themes (high-contrast, sepia/cream for dyslexia). `suppressHydrationWarning` **on** `<html>` **is required because next-themes mutates the class before React hydration.**

Content]Section B: H5P Interactive Content

Architecture]B1. H5P Core Architecture

Title: H5P — Create and Share Rich HTML5 Content Organization: H5P Group Year: 2025 (continuously updated) URL: https://h5p.org/ Summary: H5P (HTML5 Package) is a free, open-source content collaboration framework. Content

is authored in-browser, stored as `.h5p` **zip files, and rendered**

by a JavaScript runtime. It integrates with Moodle, WordPress, Drupal

via plugins, and with Canvas, Brightspace, and Blackboard via LTI.

ACESS relevance: ACESS can embed H5P content (Interactive

Videos, Quizzes, Flashcards) from its Moodle integration, leveraging

H5P's existing content library without rebuilding interactive content

types.

Title: Content Type Development — H5P Architecture

Organization: H5P Group Year: 2024 URL:

https://h5p.org/library-development Summary: H5P uses a modular

library architecture. Each content type is a library with a

`library.json` manifest, JavaScript, and CSS. Libraries can

depend on other libraries; custom content types are created by authoring

a new library with `H5P.init()`. ACESS relevance:

Explains how ACESS could create custom H5P content types for

dyslexia/ADHD-adapted interactions (e.g., reading-pace controls,

simplified drag-and-drop), or wrap existing types with accessibility

overlays.

Accessibility]B2. H5P Content Types and

Accessibility

Title: H5P Interactive Content Guide for Moodle 2026

Organization: MooDIY Blog Year: 2026 URL:

https://blog.moodiycloud.com/h5p-interactive-content-guide-moodle-2026

Summary: Catalogues H5P content types and their accessibility features. Keyboard navigation, ARIA labels, focus indicators, and high-contrast support are present across content types. Notes that Drag and Drop has a list-based keyboard fallback but differs from the visual experience. ACESS relevance: Guides ACESS educators on which H5P types are fully accessible for learners with motor impairments (avoid Drag and Drop for assessments) and which are safe for screen reader users (Quiz, Flashcards, Interactive Video).

Title: H5P and Inclusive Learning — UBC Wiki

Organization: University of British Columbia Year:

2024 URL: https://wiki.ubc.ca/Documentation:H5P

Summary: Documents how H5P supports multi-modal strategies.

Specifically notes improvements in learning for deaf and hard-of-hearing

language learners, benefits for metacognition and retrieval practice,

and UDL design guidelines for H5P authors (avoid images of text, provide

introductions). ACESS relevance: Provides evidence-based

guidance for ACESS educators authoring H5P content — avoid images of

text (screen reader incompatible), chunk content, align H5P interactions

to learning objectives.

Integration]B3. H5P in Moodle — mod_h5pactivity

Integration

Title: H5P — MoodleDocs Organization: Moodle HQ

Year: 2025 URL: https://docs.moodle.org/502/en/H5P

Summary: In Moodle, H5P content is managed via the Content

Bank. Teachers create or upload `.h5p` files, add them as H5P

activities in courses, and grades are reported.

`mod_h5pactivity` **sends xAPI statements. LUMI desktop app can**

author H5P offline. ACESS relevance: Documents the exact

pathway for ACESS's Moodle-style H5P workflow: educators upload to

Content Bank → embed in courses → learner scores feed back into ACESS's

progress tracking via xAPI.

Patterns]Section C: Moodle LMS Architecture &

Patterns

Architecture]C1. Moodle Core

Architecture

Title: Moodle Architecture — Developer Documentation

Organization: Moodle HQ Year: 2024 URL:

https://docs.moodle.org/dev/Moodle_architecture Summary:

Describes Moodle's LAMP-based architecture (Linux, Apache, MySQL, PHP)

with a pluggable subsystem: courses/activities, authentication plugins,

enrolment plugins, repository plugins. Courses are sequences of

activities organized into sections, with users enrolled in roles.

ACESS relevance: Explains the reference architecture ACESS

mirrors: course → sections → activities, with user roles

(admin/educator/learner) mapped to Moodle's manager/teacher/student

equivalents.

Title: Understanding the Moodle Architecture — Packt+

Organization: Packt Publishing Year: 2024

URL:

https://subscription.packtpub.com/book/web-development/9781801816724/2/c

Summary: The Moodle Core layer stores courses, users, roles,

groups, competencies, learning plans, and grades in a database, and

files in a separate `moodledata` directory. Covers plugin

architecture: blocks, modules, authentication, enrolment, and theme

systems. ACESS relevance: ACESS's database schema mirrors Moodle's separation of relational metadata (Supabase PostgreSQL) from file storage (Supabase Storage), with analogous plugin-style extension points.

Control]C2. Moodle Role-Based Access Control

Title: Access API — Moodle Developer Resources

Organization: Moodle HQ Year: 2025 URL: https://moodledev.io/docs/5.0/apis/subsystems/access Summary: Moodle uses a context-hierarchy RBAC model. Roles are sets of capabilities assigned in contexts (System → Course Category → Course → Module). `has_capability()` **checks permissions. Key system roles: Manager, Course Creator, Teacher, Non-editing Teacher, Student, Guest. ACESS relevance: ACESS's permission model (Admin > Educator > Learner) is a simplified version**

of Moodle's RBAC. Supabase RLS policies enforce the same isolation:

educators only see their enrolled learners; learners only see their own

progress.

Title: Role-Based Access Control with Moodle

Organization: Catalyst IT Australia Year: 2026

URL:

https://www.catalyst-au.net/blog/role-based-access-control-moodle

Summary: Explains Moodle RBAC best practices including

minimizing global roles, the "Authenticated User" base role, and

avoiding over-privileged role assignments. Covers the Principle of Least

Privilege and context-scoped role inheritance. ACESS relevance:

Validates ACESS's decision to have educators scoped to their courses

(not site-wide) and admins with elevated but audited access, rather than

a flat permission model.

Completion]C3. Moodle Course Management &

Completion

Title: Moodle Architecture — Open Source Applications

Organization: The Architecture of Open Source Applications

(aosabook.org) Year: 2023 URL:

https://aosabook.org/en/v2/moodle.html Summary: Deep technical

breakdown of Moodle's permissions system (ALLOW, PREVENT, PROHIBIT,

INHERIT), context tree, and language pack system. Explains how role

permissions aggregate at query time. ACESS relevance: Provides

the authoritative design reference for ACESS's capability-check pattern

— centralizing permission logic in database RLS policies rather than

scattered application code.

Guidelines]Section D: E-Learning Accessibility Standards &

Guidelines

Principles]D1. WCAG 2.2 — Four

Principles

Title: Web Content Accessibility Guidelines (WCAG) 2.2

Organization: W3C Web Accessibility Initiative (WAI)

Year: October 2023 (ISO/IEC 40500:2025 as of October 2025)

URL: https://www.w3.org/TR/WCAG22/ Summary: The international standard for web accessibility. Organized around four principles (POUR): Perceivable, Operable, Understandable, Robust.

Contains 13 guidelines and 86 testable success criteria at three levels (A, AA, AAA). WCAG 2.2 adds 9 new criteria over 2.1, including Focus Appearance (2.4.13), Target Size Minimum (2.5.8), and Accessible Authentication (3.3.8). ACESS relevance: ACESS targets WCAG 2.2 Level AA as its baseline, covering all learner interactions — keyboard navigation, sufficient color contrast, error identification, accessible authentication — especially critical for learners with motor and visual impairments.

Title: WCAG 2 Overview — WAI Organization: W3C Web

Accessibility Initiative (WAI) Year: 2025 URL:

https://www.w3.org/WAI/standards-guidelines/wcag/ Summary:

Overview of WCAG 2.0, 2.1, and 2.2 with links to supporting documents.

Notes backward compatibility: content conforming to WCAG 2.2 also

conforms to 2.1 and 2.0. Links to Understanding WCAG 2.2, Techniques,

and Quick Reference. ACESS relevance: The Quick Reference

(linked here) is ACESS developers' day-to-day conformance checklist

during component QA.

Practices]D2. WAI-ARIA Authoring

Practices

Title: ARIA Authoring Practices Guide (APG)

Organization: W3C Web Accessibility Initiative (WAI)

Year: 2024 (continuously updated) URL:

https://www.w3.org/WAI/ARIA/apg/ Summary: The W3C's definitive

reference for implementing accessible interactive components —

dialogs, tabs, accordions, carousels, combo boxes, tree views, and more.

Provides keyboard interaction models, required ARIA markup, and working

code examples. ACESS relevance: ACESS's custom components

(Lesson Navigation Tabs, Settings Accordion, Quiz Dialog) reference APG

patterns directly to ensure correct keyboard behavior and screen reader

announcements.

Title: APG Patterns Directory Organization: W3C Web

Accessibility Initiative (WAI) Year: 2024 URL:

https://www.w3.org/WAI/ARIA/apg/patterns/ Summary: Full catalog

of accessible patterns including Alert, Button, Carousel, Checkbox,

Combobox, Dialog (Modal), Disclosure, Feed, Grid, Listbox, Menu,

Navigation, Radiogroup, Slider, Spinbutton, Switch, Table, Tabs,

Toolbar, Tooltip, Tree View. ACESS relevance: Direct reference

for ACESS developers building any interactive component not provided by

Radix/shadcn — e.g., a custom Reading Pace Slider or a Visual Schedule

Timeline.

0.0.1 D3. Section 508

Title: Section 508 — International Harmonization

Organization: U.S. Access Board / Section508.gov Year:

2024 URL:

https://www.section508.gov/manage/laws-and-policies/international/

Summary: Section 508 of the Rehabilitation Act requires US

federal agencies to make ICT (websites, software, documents, hardware)

accessible. The 2017 refresh adopted WCAG 2.0 Level AA. Covers

harmonization with EN 301 549. ACESS relevance: If ACESS is

adopted by US federally-funded educational institutions, Section 508

compliance (WCAG 2.0 AA) is a legal requirement. Meeting WCAG 2.2 AA

exceeds this threshold.

Framework]D4. Universal Design for Learning (UDL) — CAST Framework

Title: UDL Guidelines 3.0 Organization: CAST (Center for Applied Special Technology) Year: July 2024 URL: https://udlguidelines.cast.org/ Summary: UDL 3.0 is the latest iteration, providing concrete suggestions across three networks: Engagement (why we learn — affective network), Representation (what we learn — recognition network), and Action & Expression (how we learn — strategic network). Emphasizes asset-based approaches, learner identity, and collective learning. Guides: multiple means of engagement, multiple means of representation, multiple means of action and expression. ACESS relevance: UDL is the primary instructional design framework underpinning ACESS. Font/color/spacing customization → Multiple Means of Representation. Gamification and streaks → Multiple Means of Engagement. Text-to-speech and varied response modes

→ Multiple

Means of Action & Expression.

Title: About Universal Design for Learning — CAST

Organization: CAST Year: 2024 URL:

https://www.cast.org/resources/about-universal-design-for-learning/

Summary: Explains that UDL is grounded in neuroscience — the

three brain networks model. The goal is learner agency: purposeful &

reflective, resourceful & authentic, strategic & action-oriented. UDL

is an educational framework, not an accommodation checklist.

ACESS relevance: Frames why ACESS provides learner-controlled

accessibility settings rather than admin-assigned accommodations — UDL

positions the learner as an expert on their own needs.

Standard]D5. EN 301 549 — European Accessibility

Standard

Title: EN 301 549 V3.2.1 (2021) Organization: ETSI /

CEN / CENELEC Year: 2021 URL:

https://www.etsi.org/deliver/etsi_en/301500_301599/301549/03.02.01_60/en_30

Summary: The European harmonized accessibility standard for

ICT. Includes WCAG 2.1 in full plus additional requirements for non-web

software, hardware, biometrics, real-time text, and closed system

software. Required for EAA compliance (enforceable June 2025). Version

4.1.1 (with WCAG 2.2) due 2026. ACESS relevance: If ACESS is

deployed in European educational institutions, EN 301 549 compliance is

legally required under the European Accessibility Act. Meeting WCAG 2.1

AA covers the web/software portions; additional closed-system checks may

apply.

Practices]D6. Dyslexia-Friendly Design Best

Practices

Title: Inter-Letter Spacing, Inter-Word Spacing, and Font with

Dyslexia-Friendly Features: Testing Text Readability

Organization: National Institutes of Health (PMC) /

Franceschini et al. Year: 2020 URL:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7188700/
Summary:

Peer-reviewed meta-analysis of dyslexia-friendly (DF) font research.

Finds no direct evidence that DF letterform design features improve

reading accuracy or speed on their own, but increased inter-letter and

inter-word spacing consistently aids dyslexic readers. The seminal Zorzi

et al. study showed extra-large spacing (TNR Regular enlarged by 2.5pt)

improved reading for children with dyslexia. ACESS relevance:

Validates ACESS's approach of offering adjustable line spacing

(1.5–2.0×) and letter spacing, not just font switching. OpenDyslexic

and Atkinson Hyperlegible are offered as options, but spacing controls

are the evidence-based core.

Title: Inclusive Typography: Designing for Dyslexia and

Accessibility Organization: Design Shack (editorial guide)

Year: 2025 URL:

https://designshack.net/articles/typography/inclusive-typography/

Summary: Recommends fonts with distinct letterforms (Lexend,

OpenDyslexic, Atkinson Hyperlegible), line spacing of at least 1.5× body

font size, minimum 16px body text, left-aligned text (no justification),

and avoidance of italic emphasis. ACESS relevance: Directly

informs ACESS's Dyslexia Profile settings: Atkinson Hyperlegible font,

1.6× line height, 16px minimum, cream/off-white background

(`#FDFBF4`), no justified text.

Patterns]D7. ADHD-Friendly UI Design

Patterns

Title: ADHD-Friendly Web Design: Minimizing Distractions

Organization: Bureau of Internet Accessibility (BOIA)

Year: 2024 URL:

https://www.boia.org/blog/adhd-friendly-web-design-minimizing-distractions

Summary: Documents key WCAG provisions relevant to ADHD users

— auto-playing animation must be pausable/stoppable (WCAG 2.2.2),

sufficient time for tasks (WCAG 2.2.1), predictable page behavior (WCAG

3.2.1). Recommends avoiding sticky elements and ensuring motion can be

disabled. ACESS relevance: ACESS's ADHD Profile disables

auto-play, applies Motion's `reducedMotion="always"`, shows

step-by-step progress indicators, and uses task checklists — all

aligned with WCAG provisions documented here.

Title: UI/UX for ADHD: Designing Interfaces That Actually Help

Students Organization: Din Studio Year: 2025

URL:

https://din-studio.com/ui-ux-for-adhd-designing-interfaces-that-actually-help-st

Summary: Specific patterns for ADHD-inclusive student

interfaces: task dashboards (today's tasks, in-progress, up-next),

progressive disclosure (show one step at a time), high-readability

typography, and allowing users to pause and return to tasks without losing context. Fonts, line spacing, and moderate contrast are

prioritized over animation. ACESS relevance: Informs ACESS's

ADHD-focused Lesson Mode — single-step display, persistent resume

state, visible breadcrumbs, and minimal sidebar during active learning.

Design]D8. Autism Spectrum Disorder (ASD) Inclusive

Design

Title: Designing for Autism in UX — UXPA International

Organization: UXPA International Year: 2025

URL: https://uxpa.org/designing-for-autism-in-ux/

Summary: ASD users process sensory information differently;

digital interfaces can trigger sensory overload via flashing, unexpected

popups, or inconsistent layouts. Key principles: step-by-step

instructions, predictable consistent layout, calm color palettes, and

collapsible/expandable content sections. "A clean, predictable layout

with consistent placement of elements gives users a sense of stability and familiarity." ACESS relevance: ACESS's ASD Profile

enforces consistent navigation placement, disables all animations, uses

muted neutral colors, and displays visual schedules showing the lesson

sequence before beginning.

Title: Designing Inclusive and Sensory-Friendly UX for

Neurodiverse Audiences Organization: UX Magazine Year:

2024 URL:

https://uxmag.com/articles/designing-inclusive-and-sensory-friendly-ux-for-neu

Summary: Documents evidence that autistic users benefit from

muted neutral tones (bright colors can cause discomfort), clear logical

hierarchy, and predictable navigation. Sans-serif fonts like Arial and

Open Sans are recommended. Emphasizes personalization options as best

practice. ACESS relevance: Justifies ACESS's muted color

palette in the ASD Profile and the importance of the accessibility

settings panel — individual responses vary, so offering options (not

just one preset) is key.

Accommodation]D9. Visual Impairment

Accommodation

Title: Perceivable — WCAG 2.2 Principle 1

Organization: W3C / WAI Year: 2023 URL:

https://www.w3.org/TR/WCAG22/#perceivable Summary: Perceivable

guidelines require text alternatives for non-text content (1.1),

captions for time-based media (1.2), adaptable content presentation

(1.3), and distinguishable color/contrast (1.4). Key criteria: minimum

contrast ratio 4.5:1 for normal text (AA), text resize to 200% without

loss (1.4.4). ACESS relevance: ACESS's High-Contrast Profile

meets 1.4.3 (minimum contrast AA), and text scaling is built into the

CSS with `rem`-based font sizes — doubling browser base font

size scales all text correctly.

Accommodation]D10. Hearing Impairment

Accommodation

Title: Time-Based Media — WCAG 2.2 Guideline 1.2

Organization: W3C / WAI Year: 2023 URL:

https://www.w3.org/TR/WCAG22/#time-based-media Summary:

Requires captions for live and pre-recorded synchronized media (1.2.2,

1.2.4), audio descriptions for video (1.2.5), and transcripts for

audio-only content (1.2.1). ACESS relevance: All ACESS video

lessons must have closed captions. The lesson editor includes a caption

upload field; H5P Interactive Video supports embedded caption tracks.

Accommodation]D11. Cognitive Impairment

Accommodation

Title: Cognitive Accessibility — WCAG 2.2 Understanding

Document Organization: W3C / WAI Year: 2023

URL: https://www.w3.org/WAI/cognitive/ Summary:

Cognitive accessibility guidelines include clear language (3.1),

predictable behavior (3.2), input assistance to avoid errors (3.3), and

sufficient time (2.2). WCAG 2.2 added new cognitive-focused criteria:

Accessible Authentication (3.3.7/3.3.8), Redundant Entry (3.3.7), and

Consistent Help (3.2.6). ACESS relevance: ACESS's Simplified

Mode uses short sentences, visual icons alongside text labels,

step-by-step lesson flow, and auto-saves to avoid data loss —

addressing cognitive load for all learner profiles.

Systems]Section E: Adaptive & Personalized Learning

Systems

Tracing]E1. Adaptive Learning Engines & Knowledge

Tracing

Title: Deep Knowledge Tracing and Cognitive Load Estimation for

Personalized Learning Path Generation Organization: PMC / NIH

Year: 2025 URL:

https://pmc.ncbi.nlm.nih.gov/articles/PMC12246154/ Summary:

Presents a dual-stream neural network that combines knowledge state

tracking with cognitive load estimation to generate personalized

learning paths. Achieved 87.5% prediction accuracy. Identifies

"learning sweet spots" — optimal ranges of cognitive challenge

maximizing performance and retention. ACESS relevance: Provides

the theoretical basis for ACESS's adaptive lesson sequencing —

adjusting content difficulty based on learner performance history, not

just profile-based presets.

Title: An Improved Adaptive Learning Path Recommendation Model

Driven by Real-time Learning Analytics Organization: Journal of

Computers in Education / Springer (via PMC) Year: 2022

URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC9748379/

Summary: Reviews personalized learning path recommendation

approaches including collaborative filtering, content-based filtering,

and knowledge-tracing hybrid systems. Finds that real-time learner log monitoring with adaptive resource mapping outperforms static recommendation models. ACESS relevance: Supports ACESS's design decision to log lesson interactions (time-on-task, quiz attempts, replay counts) and surface these to an adaptive recommendation layer for future lesson ordering.

Standards]E2. xAPI/SCORM Learning Analytics

Standards

Title: What Is xAPI? Tracking Learner Data — Wherever It Goes

Organization: Articulate Year: 2025 URL: https://www.articulate.com/blog/what-is-xapi/ Summary: xAPI (Experience API / Tin Can) is the successor to SCORM. Records learning "statements" in Actor–Verb–Object format (e.g., "Aliff completed Lesson 3"). Stores records in a Learning Record Store (LRS), not just an LMS. Supports offline learning, mobile, VR, and game-based learning.

ACESS relevance: ACESS's `lesson_progress` table tracks

the same data as xAPI statements — learner, action, lesson, timestamp,

score. H5P in Moodle already emits xAPI; ACESS can forward these to an

LRS for cross-platform analytics.

Title: xAPI vs SCORM — iSpring Solutions

Organization: iSpring Solutions Year: 2026

URL: https://www.ispringsolutions.com/blog/xapi-vs-scorm

Summary: Compares SCORM (tracks completion, score, time;

LMS-bound; universally supported) vs xAPI (rich behavioral statements;

platform-agnostic; requires LRS). SCORM remains dominant for formal LMS

content; xAPI is preferred for rich analytics and cross-platform

learning. ACESS relevance: ACESS supports SCORM for H5P content

compatibility with Moodle's gradebook, while also logging xAPI-style

events internally for its own analytics engine.

Education]E3. Gamification in

Education

Title: The Impact of Gamification on Learning and Instruction:

A Systematic Review Organization: ScienceDirect / Educational

Research Review Year: 2020 URL:

https://www.sciencedirect.com/science/article/abs/pii/S1747938X19301058

Summary: Systematic review of gamification empirical evidence.

Finds three positive themes: student engagement and motivation, academic

achievement, and social connectivity. The most common game elements

(points, badges, leaderboards, avatars) increase goal orientation,

persistence, and learning by repetition. ACESS relevance:

Provides evidence base for ACESS's gamification features — XP points,

lesson badges, completion streaks, and progress bars. The review

supports using multiple game elements together rather than points alone

("pointification").

Title: Examining the Effectiveness of Gamification: A

Meta-Analysis Organization: Frontiers in Psychology

Year: 2023 URL:

https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2023.1

Summary: Meta-analysis confirming moderate positive effects of

gamification on learning outcomes. Notes that design principles (PBL

triangle: Points–Badges–Leaderboards) are widely used but

"pointification" (using only points) is insufficient. Context and

learner type moderate effectiveness. ACESS relevance: Warns

ACESS against over-relying on visible leaderboards for learners with

disabilities or anxiety disorders — ACESS makes leaderboards opt-in

and emphasizes personal progress over competitive rankings.

Learning]E4. Spaced Repetition & Mastery

Learning

Title: Validating the Impact of Digital Badges on Motivation

and Academic Performance Organization: Frontiers in Education

Year: 2024 URL:

https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2024.14

Summary: Study with 95 university students found digital badges

significantly enhance intrinsic motivation across all five motivational

dimensions, with minimal impact on extrinsic motivation. No significant

difference between badge design categories. ACESS relevance:

Supports ACESS's design decision to prioritize intrinsic-motivation

badges (mastery milestones, accessibility goal achievements) over

competitive badges.

(LMS)]Section F: Role-Based Learning Management Systems

(LMS)

Control]F1. LMS Role-Based Access

Control

Title: Moodle Access API — Roles, Capabilities, and Contexts

Organization: Moodle HQ Year: 2025 URL:

https://moodledev.io/docs/5.0/apis/subsystems/access Summary:

The definitive reference for Moodle's RBAC: roles are named sets of

capabilities; capabilities are granted ALLOW/PREVENT/PROHIBIT in

contexts. Context hierarchy: System → Course Category → Course → Module.

`has_capability()` **is the central access check function.**

ACESS relevance: ACESS's three-role model

(Admin/Educator/Learner) maps cleanly onto Moodle's

Manager/Teacher/Student. The context-scoped permission model guides

ACESS's Supabase RLS policy design.

Management]F2. Course Lifecycle

Management

Title: Moodle Course Management Overview Organization:

Moodle Docs Year: 2024 URL:

https://docs.moodle.org/502/en/Course Summary: Covers course

creation, adding resources and activities, section organization,

completion tracking, and the draft → published state. Backup and restore

preserve course structure. Grading integrates with the gradebook.

ACESS relevance: ACESS mirrors the Moodle course lifecycle:

Draft → Under Review → Published → Archived. Educator workflow covers

lesson authoring (Tiptap editor), activity embedding (H5P), and progress

monitoring dashboards.

E-learning]F3. Certificate Generation in

E-learning

Title: Moodle Certificate Activity — MoodleDocs

Organization: Moodle HQ Year: 2024 URL:

https://docs.moodle.org/502/en/Certificate_activity Summary:

The Certificate activity generates PDF certificates upon course

completion, customizable with text, images, and date. Uses Moodle's

completion tracking APIs to gate certificate issuance.

relevance:ACESS

relevance: ACESS's certificate system uses a similar pattern —

triggering PDF generation (via a server-side library) when a learner's

`course_completions` row is marked complete, with learner name,

course title, and date embedded.

References]Section G: Relevant Research Papers & Academic

References

Research]G1. Dyslexia — Spacing and Font

Research

Title: Inter-Letter Spacing, Inter-Word Spacing, and Font with

Dyslexia-Friendly Features: Testing Text Readability in People with and

without Dyslexia Authors: Franceschini, S. et al.

Organization: PMC / NIH Year: 2020 URL:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7188700/ Summary:

Reviews the literature on dyslexia-friendly font features and spacing.

Key finding: no direct evidence for letterform design effects, but

extra-large inter-letter spacing (based on Zorzi et al., 2012) shows

consistent benefit. OpenDyslexic showed no improvement over Arial in

accuracy or speed in controlled studies. ACESS relevance:

Informs ACESS's evidence-based approach — prioritize spacing controls

over font switching, and offer both as learner-controlled options rather

than prescribing a single "dyslexia font."

Research]G2. ADHD — Distraction-Free Interface

Research

Title: Orchestrating Attention: Bringing Harmony to the Chaos

of Neurodivergent Learning States (AttentionGuard System)

Authors: ArXiv Preprint Year: 2026 URL:

https://arxiv.org/pdf/2602.07865 Summary: Presents

AttentionGuard — an adaptive UI system that detects ADHD learning

states (focused, drifting, hyperfocused, fatigued) and applies five UI patterns: attention-responsive chunking, state-aware verification timing, dynamic distraction reduction, progress adaptation, and re-engagement scaffolding. Validates these patterns against ADHD neuroscience literature. ACESS relevance: Provides the most current research on ADHD-adaptive UI, validating ACESS's approach of

progressive disclosure (micro-chunks), pause-and-resume, and context-aware comprehension checks.

Schedules]G3. ASD — Predictability and Visual

Schedules

Title: A DeepSeek Cross-Modal Platform for Personalized Art Education in Autism Spectrum Disorder Organization: PMC / NIH Year: 2025 URL:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12749891/ Summary:

Study with 203 participants (53 neurodivergent) demonstrates that structured, predictable environments with sensory accommodation achieve

significantly better outcomes for ASD learners. Sensory comfort ratings

4.6/5, learning satisfaction 4.3/5, NAEP score improvements 20.5% vs

8.2% for traditional methods. ACESS relevance: Empirically

supports ACESS's ASD Profile features — structured visual lesson

sequences, pre-lesson schedules, sensory accommodation settings, and

predictable navigation.

Meta-Analysis]G4. Gamification

Meta-Analysis

Title: Examining the Effectiveness of Gamification as a Tool

Promoting Teaching and Learning in Educational Settings: A Meta-Analysis

Organization: Frontiers in Psychology Year: 2023

URL:

https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2023.1

Summary: Meta-analysis of gamification effectiveness across

educational contexts. Finds overall positive but moderate effects.

Distinguishes "pointification" (trivial) from meaningful game design.

Notes that context, learner population, and design principles moderate

outcomes significantly. ACESS relevance: Grounds ACESS's

gamification design in evidence — meaningful achievements tied to

skill milestones, not arbitrary point accumulation. Disability-aware

gamification avoids competitive elements that could demotivate

vulnerable learners.

Tracing]G5. Adaptive Learning — Knowledge

Tracing

Title: TutorLLM: Customizing Learning Recommendations with

Knowledge Tracing and Retrieval-Augmented Generation

Organization: ArXiv Year: 2025 URL:

https://arxiv.org/pdf/2502.15709 Summary: Proposes combining

knowledge tracing (MLFBK model predicting learner knowledge state) with

RAG to provide personalized explanations and next-step recommendations.

Achieves 10% improvement in user satisfaction and 5% increase in quiz

scores over general LLMs. ACESS relevance: Points toward

ACESS's future adaptive engine — combining learner progress data

(knowledge tracing) with lesson content retrieval (RAG) to surface the

right explanation at the right level.

Systems]G6. Adaptive Learning — Recommendation

Systems

Title: An Improved Adaptive Learning Path Recommendation Model

Driven by Real-time Learning Analytics Organization: Journal of

Computers in Education / Springer Year: 2022 URL:

https://link.springer.com/article/10.1007/s40692-022-00250-y

Summary: Proposes real-time learner log monitoring with a

recommendation model that continuously adapts to changing learning

preferences and performance. Hybrid approach combining collaborative

filtering and knowledge tracing outperforms static models.

relevance:ACESS

relevance: Supports ACESS's roadmap for an adaptive recommendation

layer — using real-time lesson interaction logs to adjust subsequent

lesson sequencing, not just initial onboarding profiles.

Evidence]G7. Universal Design for Learning — Implementation

Evidence

Title: UDL as a Framework for Inclusion: Bridging

Neurodiversity and Learning through Sensory-Responsive Design

Organization: SENcastle Year: 2026 URL:

https://www.sencastle.com/blog/universal-design-for-learning

Summary: Reviews UDL implementation evidence. Cites CAST (2024)

UDL Guidelines 3.0. Confirms three principles (Engagement,

Representation, Action & Expression) correspond to affective,

recognition, and strategic neural networks. Evidence shows systematic

UDL implementation leads to significant improvements in educational

outcomes. ACESS relevance: Provides the theoretical and empirical foundation for ACESS's entire accessibility profile system, which operationalizes UDL's three principles into learner-controlled interface and content settings.

Engagement]G8. H5P Effectiveness — Learner Engagement

Title: H5P and Inclusive Learning at UBC Organization: University of British Columbia, Educational Technology Support Year: 2024 URL: https://wiki.ubc.ca/Documentation:H5P

Summary: Documents UBC's use of H5P in courses, noting benefits

for deaf and hard-of-hearing learners (interactive video with captions),

improved metacognition and attentional control from retrieval practice,

and reduced anxiety from learner-controlled pacing of interactive content. ACESS relevance: Confirms the value of embedding H5P in ACESS lessons for diverse learner populations, and validates the importance of learner control over interaction pacing — a key

accessibility principle.

E-Learning]G9. Accessibility Evaluation Frameworks for

E-Learning

Title: WCAG 2.2 and E-learning Accessibility Conformance

Organization: W3C / WAI and arc42 Quality Model Year:

2024–2025 URL: https://quality.arc42.org/standards/wcag-2-2

Summary: Documents that WCAG 2.2 contains 9 new criteria (added

over 2.1) particularly targeting cognitive disabilities, low vision, and

mobile users — Focus Not Obscured (2.4.11), Dragging Movements

(2.5.7), Target Size Minimum (2.5.8), and Accessible Authentication

(3.3.8). ISO/IEC 40500:2025 ratification. ACESS relevance:

Provides the specific new WCAG 2.2 criteria ACESS must address beyond

its WCAG 2.1 baseline: larger touch targets (2.5.8 — minimum 24×24 CSS

px), authentication without cognitive function tests (3.3.8), and focus

indicator visibility (2.4.11).

Principles]G10. Neurodivergent UX Design —

Principles

Title: The Principles of Neurodivergent UX Design Every Designer Should Know Organization: AccessibilityChecker.org Year: 2026 URL: https://www.accessibilitychecker.org/blog/neurodivergent-ux-design/

Summary: Synthesizes ADHD, ASD, dyslexia, and cognitive disability UX principles into actionable design rules: minimize distractions (no autoplay, no aggressive animation), clear task flows (step-by-step wizards), progressive disclosure, forgiving UI (easy undo, back navigation), and accessible typography. Notes that 15–20% of the global population is neurodivergent. ACESS relevance: Provides a combined neurodivergent UX framework that unifies ACESS's separate accessibility profiles into a coherent design system — any feature benefiting one neurodivergent group typically benefits all users under

cognitive load.

ACESS]Quick Reference: Standards Crosswalk for

ACESS

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ACESS Feature
Primary Standard
Supporting Standard

Keyboard navigation WCAG 2.2 (2.1.1) WAI-ARIA APG Patterns

Color contrast WCAG 2.2 (1.4.3 AA) EN 301 549 §9.1.4.3

Screen reader support WAI-ARIA 1.3 + Radix UI WCAG 2.2 (4.1.2)

Captions on video WCAG 2.2 (1.2.2) Section 508

Font/spacing settings Dyslexia research (Zorzi et al.) UDL

Representation

Reduced motion WCAG 2.2 (2.3.3) Motion

`useReducedMotion()`

Gamification Frontiers meta-analysis 2023 UDL Engagement

Progress tracking xAPI / SCORM Moodle `mod_h5pactivity`

Learner data isolation Supabase RLS (PostgreSQL) Moodle RBAC

principles

Educator course authoring Tiptap extensibility model Moodle course

management

Interactive content H5P (h5p.org) Moodle Content Bank

Adaptive sequencing Knowledge tracing research UDL Action &

Expression

Software Engineering Project*Document compiled: June 2026 | ACESS System — UTeM*

***Software Engineering Project* All URLs verified as of June 2026.**